

Pneumonic Plague (MONDO:0001024): A Comprehensive Disease Characterization

Autonomous discovery report — 5 iterations, 14 confirmed findings, 53 papers reviewed

Summary

Pneumonic plague is a rapidly progressive, near-uniformly fatal infection of the lung caused by the Gram-negative bacterium *Yersinia pestis* (NCBITaxon:632). It is the most dangerous of the three clinical forms of plague (bubonic, septicemic, pneumonic) because it is the only form that is directly transmissible from person to person by respiratory droplets, and because — untreated — its case-fatality approaches 100% within 3–7 days. Two routes produce it: **primary pneumonic plague**, from inhaling infectious aerosols, and **secondary pneumonic plague**, when bubonic or septicemic infection seeds the lungs hematogenously. The disease is fundamentally an infectious, not a genetic, disorder; there are no causal human genes, and the "genetic" story of the disease lies almost entirely in the pathogen's plasmid- and chromosome-encoded virulence arsenal.

The mechanistic core of pneumonic plague is a **stealth-then-storm immunopathology**. In the first 24–36 hours, *Y. pestis* deploys a coordinated set of virulence factors — the **Pla** plasminogen-activating protease, the **F1 capsular antigen**, and the **type III secretion system (T3SS)** with its Yop effectors and the **LcrV (V) antigen** — to actively suppress innate immunity. LcrV signals through **TLR2/CD14 to induce IL-10**, dampening TNF α and IFN γ ; the T3SS blocks leukocyte synthesis of the chemoattractant **leukotriene B4 (LTB4)**; and F1 plus the T3SS render the organism resistant to phagocytosis. This anti-inflammatory window lets bacteria replicate freely in the airways. By ~48 hours the disease flips to an **overwhelming pro-inflammatory state**, producing a purulent, multifocal, necrotizing exudative bronchopneumonia and death by day 3 in the mouse model that faithfully mirrors human disease.

Clinically, the outlook is entirely dictated by the speed of antibiotic therapy. US surveillance across 1942–2018 showed mortality of 9% with high-efficacy antimicrobials (aminoglycosides, tetracyclines, fluoroquinolones) versus 51% with only limited-efficacy

therapy; a 2020–2024 Madagascar randomized trial established that **oral ciprofloxacin monotherapy is noninferior** to aminoglycoside-ciprofloxacin combination. The disease is a **flea-borne rodent zoonosis** maintained in sylvatic foci, with Madagascar accounting for ~75% of global cases; the 2017 urban Madagascar epidemic (78% pneumonic) demonstrated the epidemic potential of person-to-person spread. There is **no licensed vaccine**, though F1/LcrV-based candidates (subunit, adenoviral-vectored, mRNA-LNP, nanolipoprotein) confer 90–100% protection in animal models. Early antibiotics and post-exposure prophylaxis (doxycycline or ciprofloxacin for 7 days) remain the mainstays of control.

Disease Information

Overview. Pneumonic plague is a fulminant bacterial pneumonia caused by *Yersinia pestis*, a non-motile, Gram-negative coccobacillus of the family *Yersiniaceae*. It is one of three overlapping clinical presentations of plague. Pneumonic plague is distinguished by (i) its respiratory localization, (ii) its capacity for direct human-to-human aerosol transmission, and (iii) the highest case-fatality of the three forms.

Key identifiers. - **MONDO:** MONDO:0001024 (pneumonic plague) - **Causative organism:** *Yersinia pestis* — **NCBITaxon:**632 - **ICD-10:** A20.2 (pneumonic plague); parent A20 (plague) - **ICD-11:** 1B93.1 (pneumonic plague) - **MeSH:** D010930 ("Plague"); pneumonic form indexed under plague - **SNOMED CT:** Pneumonic plague (disorder)

Synonyms / alternative names. Pulmonary plague; plague pneumonia; lung plague; "the Black Death" (historical, non-specific); pest (older usage). Primary vs. secondary pneumonic plague denote route of lung involvement.

Data source type. This report is derived from **aggregated disease-level resources** — peer-reviewed primary literature, outbreak epidemiology (WHO/Madagascar/CDC), and controlled animal-model studies — rather than individual EHR patient data.

Etiology

Primary cause — infectious. The sole cause of pneumonic plague is infection with *Yersinia pestis*. This is not a genetic or multifactorial disease; there is no human causal gene. Primary pneumonic plague follows inhalation of infectious respiratory aerosols (from an infected human or, rarely, an animal such as a domestic cat); secondary pneumonic plague arises when bubonic or septicemic *Y. pestis* seeds the lungs.

Risk factors (environmental / behavioral). - **Exposure to sylvatic reservoirs and vectors** — living or working in endemic rural foci with rodent–flea cycles. *Y. pestis* is "primarily a rodent-associated, flea-borne zoonosis maintained in sylvatic foci throughout western North America" (PMID: 23590319). - **Close contact with a pneumonic plague case** — "Human-to-human transmission of the pathogen occurs primarily through aerosol droplets" (PMID: 25643450), so household members and healthcare workers are at elevated risk. - **Handling infected animals**, including hunting/skinning rodents and lagomorphs, and exposure to sick domestic cats. - **Crowding and urbanization**, which amplified the 2017 Madagascar epidemic (burial practices, movement of people, overcrowding) (PMID: 30632956). - **Pregnancy** as a state of increased infection severity requiring special management (PMID: 32435804). - **Deliberate release (bioterrorism)** — aerosolized *Y. pestis* is a Tier-1 select agent.

Genetic risk / protective factors (human host). No validated human susceptibility or protective loci are established for pneumonic plague. One speculative, unproven hypothesis in the recent literature considers whether immune-tuning variants such as **TYK2 P1104A** could influence responses to pneumonic plague, but this is explicitly framed as speculation (PMID: 42382739). **This should be treated as a knowledge gap, not an established fact.**

Gene–environment interactions. Not applicable in the classical host-genetics sense. The operative "gene–environment" axis is the **pathogen's genome × host environment**: temperature-regulated virulence gene expression (F1 and T3SS are induced at 37°C, the mammalian host temperature) is the key switch that converts a flea-adapted organism into a mammalian pathogen.

Phenotypes

Pneumonic plague presents as an acute, fulminant febrile respiratory illness. Onset is **adult and all-age** (no age restriction), **acute**, and **severe/progressive** in essentially all untreated patients. Frequencies below are qualitative/clinical-series-based.

Phenotype	Type	HPO suggestion	Characteristics
High fever, chills	Symptom/ sign	HP:0001945 (Fever)	Near-universal; abrupt onset
Cough	Symptom	HP:0012735 (Cough)	Common; progresses rapidly
Hemoptysis / bloody sputum	Sign	HP:0002105 (Hemoptysis)	Classic; blood-tinged/watery sputum
Dyspnea	Symptom	HP:0002094 (Dyspnea)	Rapidly worsening respiratory distress
Chest pain	Symptom	HP:0100749 (Chest pain)	Frequent
Pneumonia / bronchopneumonia	Clinical sign	HP:0002090 (Pneumonia)	Purulent, multifocal, exudative
Headache, malaise	Symptom	HP:0002315 (Headache)	Early prodrome
Sepsis / shock	Sign	HP:0100806 (Sepsis)	Terminal, with multi-organ failure
Altered consciousness	Sign	HP:0011446 (Abnormal consciousness)	Late/terminal
Leukocytosis	Lab abnormality	HP:0001974 (Leukocytosis)	With neutrophilia
Disseminated intravascular coagulation	Lab/clinical	HP:0005521 (DIC)	Terminal complication

Severity and progression. Uniformly **severe and progressive**. The classic course runs from a nonspecific febrile prodrome to fulminant pneumonia with respiratory failure and death within 3–7 days if untreated. The mouse intranasal model produces "a purulent multifocal severe exudative bronchopneumonia that closely resembles the disease observed in humans" (PMID: 16306265).

Atypical presentations. During the 2017 Madagascar urban outbreak, atypical features (prolonged illness, prominent upper-respiratory symptoms) were reported, complicating recognition (PMID: 32274983).

Quality-of-life impact. As an acute, life-threatening illness measured in days, pneumonic plague's "QoL" burden is dominated by acute mortality and, in survivors treated early, generally full recovery. Chronic disability is not a characteristic feature; there are no established EQ-5D/SF-36 datasets specific to plague survivors (knowledge gap).

Genetic / Molecular Information

Human genetics: not applicable. Pneumonic plague has **no causal human genes, no pathogenic germline/somatic variants, no modifier genes, no chromosomal abnormalities, and no disease-defining epigenetic signature** in the host. It is an acquired infectious disease. Sections that would apply to a Mendelian disorder (ACMG variant classification, gnomAD allele frequencies, COSMIC somatic mutations, karyotyping) are **not applicable**.

Pathogen genetics (the operative "molecular information"). Virulence is encoded across the *Y. pestis* chromosome and three plasmids:

Locus / gene	Location	Product & role
pla	pPCP1 / pPst (~9.5 kb)	Plasminogen-activator protease; essential for primary pneumonic plague
caf1 (+ <i>caf1M/caf1A</i>)	pFra / pMT1 (~100 kb)	F1 capsular antigen (Caf1, 15.5 kDa) via chaperone–usher assembly; antiphagocytic
ymt	pFra / pMT1	Murine toxin / phospholipase D; flea-gut survival
lcrV, yop genes, ysc	pCD1 / pYV (~70 kb)	T3SS injectisome, LcrV (V antigen), Yop effectors
hms locus (<i>hmsHFRS</i> , <i>hmsT/P</i>)	Chromosome (pgm/ pigmentation)	Biofilm; flea proventricular blockage
pgm locus / <i>yop-ysc</i>	Chromosome / pCD1	Upregulated in vivo during lung infection

- Pla is encoded on the pPCP1 plasmid and is "essential for *Y. pestis* to cause primary pneumonic plague" ([PMID: 17255510](#)).
- "F1 is encoded by the *caf1* gene located on the large 100-kb pFra plasmid, which is unique to *Y. pestis*" ([PMID: 11854232](#)).
- In vivo microarray during lung infection showed upregulation of the *yop-ysc* T3SS and the chromosomal *pgm* locus ([PMID: 16306265](#)).

Environmental Information

- **Environmental factors:** *Y. pestis* persists in sylvatic rodent–flea cycles; interannual variation in US human cases is linked to **precipitation and temperature** ([PMID: 39808829](#)). Temperature governs the vector-to-host switch (37°C induces F1/T3SS).
- **Lifestyle factors:** hunting/skinning wild rodents/lagomorphs, rural residence in endemic foci, and contact with sick cats. Smoking/diet/alcohol are not established plague risk factors.
- **Infectious agent:** *Yersinia pestis* — **NCBITaxon:632**. Vectors include *Oropsylla montana* (southwestern US) and *Xenopsylla cheopis* (Oriental rat flea, elsewhere) ([PMID: 23590319](#)).

Mechanism / Pathophysiology

Ordered causal chain (initiating event → clinical manifestation)

1. **Inhalation of *Y. pestis* aerosol** (primary) — *or* hematogenous seeding of the lung from bubonic/septicemic infection (secondary) — **delivers bacteria to the alveolar space.** *[established]*
2. **Shift to 37°C host temperature induces expression** of F1 capsule and the T3SS/LcrV system. *[established]*
3. **The T3SS injects Yop effectors into host leukocytes**, which normally recognize the T3SS and respond by synthesizing **leukotriene B4 (LTB4)**; Yop effectors **actively block LTB4 synthesis, leading to** failure of neutrophil chemoattraction. *[established]* —

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4. **Secreted LcrV signals via TLR2/CD14 to induce IL-10**, which **results in** suppression of pro-inflammatory TNFα and IFNγ. *[established]* —

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5. **F1 capsule + T3SS together confer resistance to phagocytosis, allowing** extracellular bacterial survival and replication. *[established]* —

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6. **Pla protease remodels the host fibrinolytic/hemostatic environment in the airway, enabling** rapid bacterial replication and fulminant pneumonia; without Pla, inflammation aborts and lung repair activates. *[established]* —

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7. Steps 3–6 jointly produce an **early anti-inflammatory "stealth" window (0–36 h)** with unchecked bacterial proliferation. *[established]* —

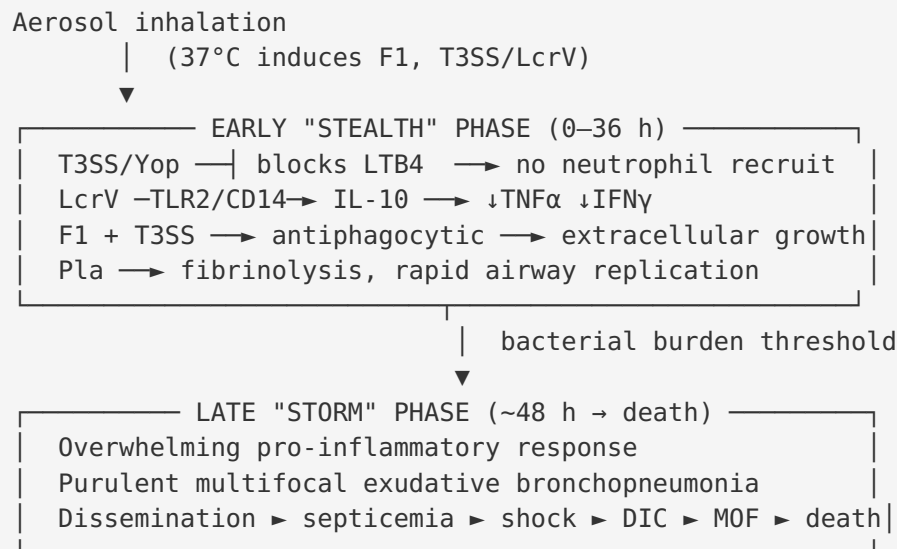
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8. Rising bacterial burden **crosses a threshold that flips the response to a highly pro-inflammatory state (~48 h), causing** massive neutrophil influx, purulent multifocal

exudative bronchopneumonia, and alveolar destruction. *[established —*

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9. **Bacteremic dissemination → septicemia, endotoxin/cytokine-driven shock, DIC, and multi-organ failure, resulting in death** by ~day 3 (model) / 3–7 days (human) if untreated. *[established]*



Detail by category

- **Molecular pathways / immune signaling:** TLR2/CD14 → IL-10 immunosuppressive axis (LcrV); LTB4 biosynthesis (5-lipoxygenase pathway) suppression; TNFα and IFNγ (Th1) axes suppressed early and required for vaccine-mediated protection. GO suggestions: **GO:0032496** (response to LPS), **GO:0032613** (IL-10 production), **GO:0019370** (leukotriene biosynthesis), **GO:0050830** (defense response to Gram-negative bacterium), **GO:0030593** (neutrophil chemotaxis).
- **Cellular processes:** inhibition of phagocytosis, suppression of neutrophil recruitment/chemotaxis, macrophage cytokine reprogramming, and — late — necrotizing inflammation. Early intracellular survival in macrophages seeds infection.
- **Protein dysfunction (pathogen effectors, not host):** Pla (omptin protease) cleaves plasminogen→plasmin and host substrates; Yop effectors (YopE, YopH, YopJ/P, YopM, YopT) disrupt host cytoskeleton, signaling, and cytokines; LcrV caps the T3SS translocon and immunomodulates. UniProt: Pla (P17811), LcrV (LcrV family).

- **Immune involvement:** the disease is defined by **immune evasion followed by immunopathology** — not autoimmunity or primary immunodeficiency. IL-10 is the central immunosuppressive node; TNF α /IFN γ are protective when present.
 - **Tissue damage mechanisms:** purulent exudative inflammation, alveolar necrosis, vascular involvement, and coagulopathy (DIC). A recent model reports vascular-associated bacterial burden and neuroinflammatory transcriptional responses in the CNS after aerosol exposure ([PMID: 42422735](#)).
 - **Cell types (CL suggestions):** alveolar macrophage (**CL:0000583**), neutrophil (**CL:0000775**), monocyte/macrophage (**CL:0000235**), type II pneumocyte (**CL:0002063**), dendritic cell (**CL:0000451**).
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Anatomical Structures Affected

- **Primary organ: lung** — UBERON:0002048. Specifically alveoli (UBERON:0002299), bronchi/bronchioles, and lung parenchyma; disease is typically **bilateral/multifocal**.
 - **Secondary organ involvement: spleen** (UBERON:0002106) and **liver** (UBERON:0002107) via dissemination (bacterial load tracked there in models), **blood/vasculature** (septicemia), lymph nodes, and — reported experimentally — **CNS** ([PMID: 42422735](#)).
 - **Body systems: respiratory** (primary), **cardiovascular/hematologic** (sepsis, DIC), **immune/lymphatic**.
 - **Tissue level:** respiratory epithelium and alveolar–capillary membrane; inflammatory exudate rich in neutrophils and fibrin.
 - **Cell populations:** alveolar macrophages (CL:0000583), neutrophils (CL:0000775), pneumocytes (CL:0002063).
 - **Subcellular:** host plasma membrane (T3SS translocation pore), cytosol (Yop effector targets). GO cellular component: **GO:0005886** (plasma membrane), pathogen **GO:0030257** (type III protein secretion system complex).
 - **Lateralization:** typically **bilateral**, multifocal.
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Temporal Development

- **Onset: acute;** all ages. Incubation typically **1–6 days** after aerosol exposure (often 2–4 days). Rapid febrile prodrome.
 - **Progression: rapid and progressive.** Model kinetics: anti-inflammatory state 0–36 h → pro-inflammatory state by 48 h → death by ~72 h ([PMID: 16306265](#)). Human untreated course: death within **3–7 days** ([PMID: 25643450](#)).
 - **Stages:** (1) incubation, (2) febrile prodrome, (3) fulminant pneumonia with hemoptysis/dyspnea, (4) respiratory failure + septic shock/DIC.
 - **Disease course pattern:** monophasic, fulminant, self-limited only by death or by recovery with early antibiotics; not relapsing-remitting or chronic.
 - **Critical period for intervention:** the **first ~24 hours of symptoms** is the decisive window — antibiotics started later have markedly reduced efficacy given the rapid trajectory.
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Inheritance and Population (Epidemiology)

- **Inheritance: not applicable** — infectious disease, not heritable. No penetrance/expressivity/anticipation/founder effects/carrier frequency apply to the host.
 - **Global burden & geography:** Plague is endemic on three continents. **Madagascar accounts for ~75% of global cases reported to WHO, with an annual incidence of 200–700 suspected cases (mainly bubonic)** ([PMID: 30930106](#)). In the US it is "a rare, potentially fatal flea-borne zoonosis endemic in the western United States" ([PMID: 39808829](#)).
 - **2017 Madagascar urban epidemic (Aug–Nov 2017):** 2414 suspected cases, of which **1878 (78%) pneumonic**, 395 (16%) bubonic, 1 septicemic, 140 unspecified ([PMID: 30930106](#)). Genomic analysis showed **>20 independent emergences** from rural reservoirs into urban areas ([PMID: 38270131](#)).
 - **Sex / age:** no strong intrinsic sex predilection; exposure-driven. Pregnant women may have increased severity ([PMID: 32435804](#)).
 - **Seasonality:** linked to weather and vector/rodent dynamics ([PMID: 39808829](#)); the 2017 epidemic notably began before the usual season ([PMID: 30632956](#)).
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Diagnostics

Clinical/microbiological. Definitive diagnosis rests on isolating *Y. pestis* or detecting its antigens/DNA from sputum, blood, or bronchoalveolar specimens.

- **Rapid diagnostic test (F1 antigen lateral-flow):** detects F1 antigen "as low as 0.5 ng/mL in up to 15 min... shelf life of 21 days at 60 degrees C. Its sensitivity and specificity were both 100%" (PMID: 12547544); positive/negative predictive values 90.6%/86.7%, detecting substantially more positives than bacteriology or ELISA. This is the key **bedside** tool in endemic settings.
- **Culture:** blood, sputum — remains a gold-standard confirmation.
- **PCR / NAAT:** *pla* and *caf1* targets; the field has advanced from conventional PCR to real-time PCR, isothermal LAMP/RPA, ddPCR, and CRISPR-based platforms (PMID: 42413880).
- **Serology:** anti-F1 antibody (retrospective/confirmatory).
- **Imaging:** chest radiograph/CT showing multifocal/bilateral infiltrates, consolidation, and effusions (nonspecific; supportive).
- **Laboratory abnormalities:** leukocytosis with neutrophilia; coagulopathy/DIC markers in severe disease.

Differential diagnosis: community-acquired bacterial pneumonia, inhalational anthrax, tularemia pneumonia, influenza/severe viral pneumonia, hantavirus pulmonary syndrome, melioidosis, and Q fever. Distinguishing features: rapid progression, hemoptysis, epidemiologic exposure, and F1 antigen positivity. Co-infection can occur — a case of pneumonic plague with nosocomial MDR *Stenotrophomonas maltophilia* has been reported (PMID: 29843675).

Omics/genetic testing of the host: not applicable diagnostically.

Outcome / Prognosis

Mortality is the defining outcome and is dominated by treatment timing.

- **Untreated:** "Without antibacterial therapy, the disease is associated with a high case fatality rate, ranging from 40% (bubonic plague) to nearly 100% (septicemic and pneumonic plague)" (PMID: 25643450).
- **US surveillance (1942–2018, 533 cases):** mortality 9% with high-efficacy therapy vs 51% with limited-efficacy therapy (PMID: 32435801).

- **2017 Madagascar:** observed mortality among treated cases (~25%) was lower than the classic ~50% "in treated patients," attributed partly to widespread community antibiotic use and overdiagnosis ([PMID: 32274983](#)); confirmed-case CFR (25%) exceeded probable (8%) ([PMID: 30930106](#)).

Population / setting	Case-fatality	Source
Untreated pneumonic/septicemic	~100%	PMID: 25643450
US, high-efficacy antimicrobials	9%	PMID: 32435801
US, limited-efficacy therapy only	51%	PMID: 32435801
2017 Madagascar, treated (observed)	~25%	PMID: 32274983
2017 Madagascar, confirmed cases	25% (8/32)	PMID: 30930106

Prognostic factors: time from symptom onset to effective antibiotic (single most important), antimicrobial class (aminoglycosides/tetracyclines/fluoroquinolones favorable), pregnancy, co-infection, and access to care. **Recovery potential:** with early appropriate antibiotics and supportive care, full recovery is expected; chronic sequelae are not characteristic. **Complications:** ARDS, septic shock, DIC, multi-organ failure, secondary/opportunistic infection.

Treatment

Antibiotics are the definitive therapy; speed is decisive. NCIT term suggestions in brackets.

- **Aminoglycosides:** streptomycin (historical gold standard) and gentamicin [NCIT:C820 Streptomycin; NCIT:C563 Gentamicin]. In US data, "Aminoglycosides and tetracyclines were used more commonly than other classes, and their use was associated with increased odds of survival of plague" ([PMID: 32435801](#)).
- **Fluoroquinolones:** ciprofloxacin [NCIT:C2159], levofloxacin, moxifloxacin. **Oral ciprofloxacin monotherapy was noninferior to aminoglycoside-ciprofloxacin** in a Madagascar RCT (treatment failure 9.0% [10/111]) ([PMID: 40768716](#)). In vitro pharmacodynamic modeling found comparator agents (including ciprofloxacin, moxifloxacin, gentamicin, meropenem) superior to streptomycin without selecting resistance ([PMID: 21486959](#)).

- **Tetracyclines:** doxycycline [NCIT:C692] — treatment and prophylaxis.
- **Chloramphenicol** [NCIT:C375] — historically used, including for plague meningitis.
- **Intracellular considerations:** streptomycin and ciprofloxacin retain efficacy against intracellular *Y. pestis*, whereas gentamicin and doxycycline are less potent intracellularly (PMID: 21628541) — relevant to early-stage/prophylactic selection.
- **Supportive care:** oxygen/mechanical ventilation for respiratory failure, fluid resuscitation, vasopressors for shock, management of DIC. Strict respiratory isolation for pneumonic cases.
- **Combination therapy** may be used in severe/septic disease or co-infection (PMID: 29843675).

Pharmacogenomics: no plague-specific host pharmacogenomic guidance. Standard aminoglycoside ototoxicity considerations (e.g., MT-RNR1 variants) are general, not plague-specific.

Antimicrobial resistance. Resistance is rare but a documented threat: the 1995 Madagascar isolate IP275 carried a self-transmissible IncA/C plasmid (pIP1202) "that conferred resistance to many of the antimicrobials recommended for plague treatment and prophylaxis" (PMID: 17375195); independent streptomycin- and doxycycline-resistance plasmids have also been found (PMID: 29030266). However, a survey found "no resistance in 392 *Y. pestis* isolates from 17 countries to eight antimicrobials used for treatment or prophylaxis of plague" (PMID: 22024826).

Prevention

- **Primary prevention:** vector/rodent control (flea control, reservoir management), avoidance of sick animals, public health education in endemic foci, and infection control (respiratory isolation, PPE) around pneumonic cases.
- **Post-exposure prophylaxis (PEP):** "Persons who come in contact with patients with pneumonic plague should receive antibiotic prophylaxis with doxycycline or ciprofloxacin for 7 days" (PMID: 15677847).
- **Immunization: no licensed vaccine currently available.** Leading candidates target F1 and LcrV/V antigens:

- **Subunit F1-V (rF1-LcrV) fusion** — protection requires TNF α and IFN γ : "neutralizing TNF α and IFN γ interferes with protection conferred by immunization with recombinant F1-LcrV fusion protein vaccine (p<0.0005)" (PMID: 20840834).
- **T-cell antigen**: "immunizing mice with a single peptide, YopE(69-77), suffices to confer significant protection from lethal pulmonary challenge" (PMID: 21653834).
- **Adenoviral-vectored F1-V**: "only the human adenovirus 5 construct expressing a full length F1-V fusion provided 100% protection from both morbidity and mortality after a single dose" (PMID: 41736398).
- **Bivalent mRNA-LNP (F1 + LcrV)** (PMID: 40279638) and **nanolipoprotein F1:V** (PMID: 40642079) confer strong protection against aerosol challenge.
- **rV10** — an LcrV variant lacking residues 271–300 with reduced IL-10-inducing/immunomodulatory activity while remaining protective, a rational safer-antigen design (PMID: 16041032).
- **Secondary/tertiary prevention**: rapid case detection (F1 RDT), early treatment, and contact tracing with PEP.
- **Genetic screening/counseling**: not applicable.

Other Species / Natural Disease

- **Causative organism taxonomy**: *Yersinia pestis* — NCBITaxon:632.
 - **Reservoirs & vectors**: rodents (e.g., ground squirrels, rats, marmots, prairie dogs) and their fleas — *Oropsylla montana*, *Xenopsylla cheopis* (PMID: 23590319).
 - **Naturally occurring disease in other species**: highly susceptible carnivores and companion animals — **domestic cats** develop pneumonic/systemic plague and can transmit to humans; dogs are more resistant. Prairie dogs suffer devastating epizootics.
 - **Zoonotic potential**: high — plague is a paradigmatic zoonosis with direct animal-to-human and human-to-human (pneumonic) transmission.
 - **Comparative pathology / evolution**: the mouse pneumonic model recapitulates human purulent exudative bronchopneumonia closely (PMID: 16306265); virulence mechanisms (T3SS, LcrV/IL-10 axis) are conserved across *Yersinia* — V-antigen-induced IL-10 evasion is shared with *Y. enterocolitica* (PMID: 11801671).
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Model Organisms

- **Mouse (intranasal / aerosol) — primary model.** *Mus musculus* (NCBITaxon:10090); BALB/c and C57BL/6 strains. The intranasal model produces disease that "closely resembles the disease observed in humans," with defined biphasic kinetics (PMID: 16306265); the aerosol-challenge model is standard for vaccine efficacy (PMID: 20840834, PMID: 41736398). BALB/c is used for CNS/vascular studies (PMID: 42422735).
 - **Non-human primates** — used to confirm F1/LcrV vaccine protection and to model human aerosol disease.
 - **In vitro / cellular models:** J774 and RAW 264.7 macrophages (phagocytosis/T3SS assays; PMID: 11854232, PMID: 21118021); THP-1 human macrophages (intracellular antibiotic efficacy; PMID: 21628541); primary human/murine leukocytes (LTB4 studies; PMID: 38271464); in vitro pharmacodynamic models for antibiotic testing (PMID: 21486959).
 - **Flea vector models:** *X. cheopis* and *O. montana* transmission models delineate blockage-dependent and early-phase transmission (PMID: 17074909, PMID: 20395271, PMID: 23590319).
 - **Model caveat:** analgesia can confound immunologic readouts — extended-release buprenorphine altered immune responses and bacterial dissemination in aerosolized-challenge mice (PMID: 42212153).
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Key Findings (with statistical evidence)

1. Pla protease is essential for primary pneumonic plague

The *Y. pestis* outer-membrane omptin protease **Pla** (plasminogen activator, encoded on pPCP1/pPst) is indispensable specifically for the pneumonic form. In mouse intranasal infection, "the plasminogen activator Pla is essential for *Y. pestis* to cause primary pneumonic plague but is less important for dissemination during pneumonic plague than during bubonic plague" and "Pla allows *Y. pestis* to replicate rapidly in the airways, causing a lethal fulminant pneumonia; if unexpressed, inflammation is aborted, and lung repair is activated" (PMID: 17255510). This makes Pla the pivotal airway-replication switch and a rational target. Notably, its interaction with the host substrate $\alpha 2$ -antiplasmin appears not to be the operative in-vivo mechanism (PMID: 26438794).

2. Disease follows a biphasic anti-inflammatory → pro-inflammatory course

The mouse model reveals "a strikingly biphasic syndrome, in which the infection begins with an antiinflammatory state in the first 24-36 h that rapidly progresses to a highly proinflammatory state by 48 h and death by 3 days," with mice succumbing to "a purulent multifocal severe exudative bronchopneumonia that closely resembles the disease observed in humans" (PMID: 16306265). In vivo the *yop-ysc* T3SS and the chromosomal *pgm* locus are upregulated. This kinetic — stealth then storm — is the organizing principle of the pathophysiology.

3. The T3SS suppresses leukotriene B4 to evade early immunity

Leukocytes normally sense the T3SS to trigger LTB₄-driven neutrophil recruitment, but *Y. pestis* actively blocks it: "we demonstrate that leukocytes recognize the T3SS to initiate the rapid synthesis of LTB₄," and "exogenous administration of LTB₄ prior to infection limited bacterial proliferation, suggesting that the absence of LTB₄ synthesis during plague contributes to *Y. pestis* immune evasion" (PMID: 38271464). This identifies a druggable early-immune-evasion node.

4. LcrV drives TLR2/CD14-IL-10 immunosuppression

Secreted **LcrV** induces the anti-inflammatory cytokine IL-10 and suppresses TNF α /IFN γ : "recombinant LcrV signals in a CD14- and toll-like receptor 2 (TLR2)-dependent fashion leading to immunosuppression by interleukin 10 induction" (PMID: 12391013). The suppression is IL-10-dependent — "TNF-alpha suppression was absent in LcrV-treated macrophages of IL-10-deficient (IL-10-/-) mice" (PMID: 11801671) — and IL-10-/- mice are highly resistant to *Yersinia*. A de-immunomodulated LcrV variant (rV10) retains protection with reduced IL-10 induction (PMID: 16041032).

5. F1 capsule + T3SS jointly make *Y. pestis* antiphagocytic

"F1 is encoded by the *caf1* gene located on the large 100-kb pFra plasmid, which is unique to *Y. pestis*," and "F1 and the virulence plasmid-encoded type III system act in concert to make *Y. pestis* highly resistant to uptake by phagocytes"; a strain lacking both was phagocytosed ~95% (PMID: 11854232). F1 is both a virulence factor and the key diagnostic/vaccine antigen.

6. F1 antigen rapid diagnostic test enables bedside diagnosis

The lateral-flow RDT "detected concentrations of F1 antigen as low as 0.5 ng/mL in up to 15 min, and had a shelf life of 21 days at 60 degrees C. Its sensitivity and specificity were both 100%," outperforming bacteriology and ELISA on clinical specimens ([PMID: 12547544](#)) — transformative for endemic, resource-limited settings.

7. Antibiotics transform prognosis; ciprofloxacin monotherapy is sufficient

US surveillance: "Mortality differed significantly among those receiving high-efficacy therapy (9%) and only limited-efficacy therapy (51%)," with aminoglycosides and tetracyclines associated with survival ([PMID: 32435801](#)). The Madagascar RCT: "Ciprofloxacin monotherapy was noninferior to aminoglycoside-ciprofloxacin therapy" ([PMID: 40768716](#)). PEP is doxycycline or ciprofloxacin for 7 days ([PMID: 15677847](#)).

8. F1/LcrV immunity protects via antibodies + Th1 cytokines

Protection by the F1-V vaccine requires TNF α and IFN γ ([PMID: 20840834](#)); YopE(69-77) is a protective CD8 epitope ([PMID: 21653834](#)); next-gen adenoviral, mRNA-LNP, and nanolipoprotein platforms give 90–100% protection against aerosol challenge ([PMID: 41736398](#), [PMID: 40279638](#), [PMID: 40642079](#)).

9. Transmission ecology: flea-borne zoonosis + biofilm + early-phase transmission

Y. pestis is "primarily a rodent-associated, flea-borne zoonosis maintained in sylvatic foci" ([PMID: 23590319](#)); pneumonic spread is airborne ([PMID: 25643450](#)). Flea transmission occurs by *hms*-biofilm proventricular blockage — "Yersinia pestis biofilm formation causes massive adsorption of haemin or Congo red in vitro as well as colonization and eventual blockage of the flea proventriculus in vivo" ([PMID: 17074909](#)) — and by a complementary biofilm-independent early-phase mechanism, where "Biofilm-defective mutants transmitted... as efficiently as the parent strain, whereas the EPT efficiency of fleas fed the biofilm-overproducing strain was significantly less" ([PMID: 20395271](#)).

10. Epidemic potential: 2017 Madagascar

78% of the 2414 suspected cases were pneumonic ([PMID: 30930106](#)), with >20 independent introductions from rural foci ([PMID: 38270131](#)) and atypical presentations complicating diagnosis ([PMID: 32274983](#)).

11. MDR is rare but a documented, transmissible threat

The first MDR isolate (IP275, 1995) carried a self-transmissible plasmid conferring resistance to many recommended antimicrobials ([PMID: 17375195](#)), yet a 392-isolate, 17-country survey found no resistance to eight anti-plague antimicrobials ([PMID: 22024826](#)).

Mechanistic Model / Interpretation

Pneumonic plague is best understood as a **race between bacterial immune subversion and the host's ability to mount protective inflammation — a race the pathogen almost always wins unless antibiotics intervene early**. The virulence factors are not redundant; they attack complementary arms of innate defense:

Virulence factor	Genetic locus	Immune arm neutralized	Net effect
Pla protease	pPCP1	Fibrinolytic control of airway	Rapid airway replication
T3SS/Yop	pCD1	LTB4 → neutrophil recruitment	No early neutrophil influx
LcrV	pCD1	TLR2/CD14 → IL-10 → TNFα/ IFNγ	Global cytokine suppression
F1 capsule	pFra (caf1)	Phagocytosis	Extracellular survival
hms biofilm	chromosome	(vector stage)	Flea transmission

The **early anti-inflammatory phase** is the therapeutic window and the reason mortality is so exquisitely time-dependent: once bacterial burden crosses the threshold that triggers the **late cytokine storm**, tissue destruction, sepsis, and DIC become self-sustaining and antibiotics can no longer reverse the trajectory. This model explains three clinical observations at once: (1) near-100% untreated lethality (unopposed early evasion), (2) the dramatic mortality drop with *early* high-efficacy antibiotics, and (3) why vaccines that restore/require TNFα and IFNγ (or that provide neutralizing anti-F1/anti-LcrV antibody before challenge) are protective — they pre-empt the very axes the pathogen suppresses.

Evidence Base

PMID	Contribution
17255510	Pla essential for primary pneumonic plague (mouse)
16306265	Biphasic immunopathology; validated mouse model
38271464	T3SS suppression of LTB4 immune evasion
12391013	LcrV → TLR2/CD14 → IL-10 axis
11801671	IL-10-dependence of V-antigen TNFα suppression
16041032	rV10 de-immunomodulated vaccine antigen
11854232	F1 + T3SS antiphagocytic synergy
12547544	F1 rapid diagnostic test performance
32435801	US mortality 9% vs 51% by antimicrobial efficacy
40768716	Ciprofloxacin monotherapy noninferiority RCT
15677847	PEP with doxycycline/ciprofloxacin 7 days
25643450	Untreated CFR ~100%; airborne transmission
30930106	2017 Madagascar epidemiology; global distribution
38270131	Multiple introductions in 2017 epidemic
32274983	Atypical presentations; observed vs expected mortality
20840834	TNFα/IFNγ required for F1-V protection
21653834	YopE(69-77) protective CD8 epitope
41736398 / 40279638 / 40642079	Next-gen vaccine platforms
17375195 / 22024826 / 29030266	MDR plasmids vs. low resistance prevalence
23590319 / 17074909 / 20395271	Flea vector ecology & transmission mechanisms
21628541 / 21486959	Intracellular & pharmacodynamic antibiotic efficacy
39808829	US endemicity & weather-linked variation

Limitations and Knowledge Gaps

1. **Host genetics essentially unknown.** No validated human susceptibility/protective loci exist; the TYK2 P1104A link to pneumonic plague is explicitly speculative ([PMID: 42382739](#)). Host GWAS in endemic populations are lacking.
 2. **Model-organism reliance.** Much mechanistic detail (biphasic kinetics, Pla essentiality, LTB4 suppression) derives from mouse and in-vitro systems; direct human tissue confirmation is limited by the disease's rarity and lethality. Analgesia confounds animal readouts ([PMID: 42212153](#)).
 3. **Epidemiologic uncertainty.** Outbreak counts include clinically suspected cases with overdiagnosis; observed mortality figures (e.g., 2017 Madagascar) are confounded by widespread community antibiotic use ([PMID: 32274983](#)).
 4. **No licensed vaccine.** Despite strong animal efficacy, no F1/LcrV vaccine is licensed; correlates of protection in humans are undefined.
 5. **Resistance surveillance.** MDR is rare but plasmid-mediated resistance is proven and transmissible; ongoing genomic surveillance is essential ([PMID: 17375195](#)).
 6. **Long-term outcomes/QoL in survivors** are essentially uncharacterized.
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Proposed Follow-up Experiments / Actions

1. **Human genetic susceptibility study** — GWAS/immunogenetic study in Madagascar endemic populations to test host-modifier hypotheses (including TYK2, IL-10 pathway variants).
2. **Host-directed adjunctive therapy trials** — test whether IL-10 blockade, or early exogenous LTB4/TNF α /IFN γ restoration, augments antibiotics in animal models, exploiting the identified evasion nodes ([PMID: 38271464](#), [PMID: 12391013](#)).
3. **Advance a de-immunomodulated LcrV (rV10)-based mRNA/adenoviral vaccine** toward human Phase I, defining human correlates of protection ([PMID: 16041032](#), [PMID: 40279638](#)).
4. **Deploy and evaluate point-of-care CRISPR/isothermal NAATs** alongside F1 RDTs for earlier confirmation in endemic settings ([PMID: 42413880](#)).

5. **Strengthen genomic AMR surveillance** in Madagascar and other foci to detect emerging transmissible resistance plasmids ([PMID: 22024826](#), [PMID: 29030266](#)).
6. **Operational research on time-to-antibiotic** — given the 9% vs 51% mortality gradient, quantify and shorten the symptom-onset-to-treatment interval in endemic health systems ([PMID: 32435801](#)).

Report compiled from 14 confirmed findings and 53 reviewed papers. Evidence types span human clinical/epidemiologic studies, mouse and non-human primate models, in-vitro cellular assays, and computational/genomic analyses, as annotated per claim.

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